

Visual Quality Robustness  
(motion blur, low-lighting perturbations)

| Knowledge-based Robustness                                                        |                                                                                                                                                                                        |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                            |
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| General Object Understanding                                                      |                                                                                                                                                                                        | Stand-alone Anomaly Detection                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                   | Pair-wise Anomaly Detection                                                                                                                                                                |
| Unanswerable or Ill-posed Query                                                   |                                                                                                                                                                                        |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                            |
|  |                                                                                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                   |                      |
| Multiple-choice                                                                   | What is the object in the image?                                                                                                                                                       | Is there a defect on the object? If yes, what is the defect and where is it located?                                                                                                                                            | Compare the query object to the defect-free reference and assess whether it has any defects. If so, identify the defect(s) and their potential effects on the object.                                                                                                             | What colors are present on the fire extinguisher in the image?                                                                                                                             |
|                                                                                   | <p>A) <b>Two-toned medicinal capsule</b></p> <p>B) E-cigarette cartridge</p> <p>C) Compressed round tablet</p> <p>D) Plastic spinning toy</p> <p>E) Small cylindrical battery cell</p> | <p>A) Object is normal, no major defects</p> <p>B) <b>Crack at center-left on black half</b></p> <p>C) Deep scratch on black half near center</p> <p>D) Hole on upper-right</p> <p>E) Faded printing over 500 on green half</p> | <p>A) Object is normal, no major defects</p> <p>B) Smudged printing, no structural damage</p> <p>C) Surface scratch at right, minor aesthetic damage</p> <p>D) Hole at right end causing capsule rupture</p> <p>E) <b>Crack on black half, cause medicine content leakage</b></p> | <p>A) Purple and orange</p> <p>B) Red, white, and black</p> <p>C) Black and orange</p> <p>D) <b>Image does not show a fire extinguisher. It is a capsule.</b></p> <p>E) Red and yellow</p> |
| Open-ended                                                                        | What is the object shown, and what are its key features (e.g. color, material, and shape)?                                                                                             | Does the object in the image have any defects? If so, describe the defect or anomaly and specify its location (e.g., upper-left, center, bottom-right).                                                                         | Compare the query to the defect-free reference, and assess whether the query object is normal or defective. If defective, describe any major anomalies with location (e.g., upper-left, center, bottom-right).                                                                    | Does the chair in the image have any defects? If so, describe the defect or anomaly and specify its location (e.g., upper-left, center, bottom-right).                                     |
|                                                                                   | List the main attributes of the product in the image (e.g., color, texture, and material).                                                                                             | Assess whether the product has any major defects. If so, how might the defect affect the object's normal function or safety?                                                                                                    | Compare the query object to the defect-free reference and assess whether it has any major defects. If so, how might the defect affect the object's normal function or safety?                                                                                                     | List the main attributes of the cardboard box in the image (e.g., color, physical condition, and material).                                                                                |
|                                                                                   | Explain the intended function or use of the object shown when it is in ideal condition.                                                                                                | Evaluate whether the product has any visible defects. If so, what distinguishes the damaged part from the rest of the object?                                                                                                   | Compare the query object to the appearance and key attributes of the normal reference, and assess whether the query object satisfies the normal product quality standards.                                                                                                        | Product appears to be normal and defect-free. Please confirm that it meets the normal product quality standards.                                                                           |